Ans. (c) : A fluorescent lamp or fluorescent tube is a gas discharge lamp that uses electricity to excite mercury vapour. The excited mercury atoms produce shortwave ultraviolet light that then causes a phosphor to fluoresce, producing visible light .In fluorescent tubes the simplest ballast for alternating current use is an inductor (choke coil) placed in series, consisting of a winding on a laminated magnetic core. The inductance of this winding limits the flow of AC current on account of its large impedance. The only waste of energy is due to the hysteresis loss in the iron core, which is much less than the waste of energy in comparison to resistance which can also reduce the current if placed instead of the choking coil.

80. For reproducing sound, a CD (Compact Disc) audio player uses a

(a) Quartz crystal
(b) Titanium needle
(c) Laser beam
(d) Barium titanate ceramic

Ans. (c) : In compact disc digital data is scanned by laser beam for the reproduction of recorded sound.

81. When a CD (compact disc used in audio and video systems) is seen in sunlight, rainbow like colours are seen. This can be explained on the basis of the phenomenon of:

(a) Reflection and diffraction
(b) Reflection and transmission
(c) Diffraction and transmission
(d) Refraction, diffraction and transmission

Ans. (d) : The reason CDs reflect rainbow colors is because they have a clear plastic coating on top of a mirrorized surface. Light refracts (bends) when it moves from one medium (such as air) to another with a different optical density (such as the clear plastic surface of a CD). Different wavelengths of light (every color has a different wavelength) travel at different speeds, so that full spectrum appears when white light passes from the air through the plastic surface of a CD, separated light rays which are then reflected back to us by the mirrorized center surface of a CD. Here the diffraction and transmission also takes place because diffraction of light rays occurs when it strikes the surface of CD and transmission is obvious when light enters from one medium to another. The thickness of the different optical media, angle of source light, and brightness of source light reflects rainbow pattern on a CD.

82. Assertion A : A man standing on a completely frictionless surface can propel himself by whistling.

Reason R : If no external force acts on a system, its momentum cannot change.

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Ans. (b) : First statement can be explained on the laws of motion. First statement involves the use of Newton.
An initial force is required to start and continue a motion at a frictionless surface. Therefore, a man standing on a completely frictionless surface can propel himself by whistling. As per the Newton's second law of motion, The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction in which the force acts. Hence, if no external force acts on a system, its momentum cannot change.

83. Assertion A : Small glass beads fixed on traffic signals glow brightly when light falls upon them.

Reason R : Light is totally reflected when the angle of incidence exceeds a certain critical value and light travelling in a denser medium is reflected from a rarer medium.

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Ans. (a) : Both the statements are correct and statement 2 is also correct explanation for statement 1. Total internal reflection is an optical phenomenon that happens when a ray of light strikes a medium boundary at an angle larger than a particular critical angle with respect to the normal to the surface. If the refractive index is lower on the other side of the boundary, no light can pass through and all of the light is reflected. Thus because of this phenomena traffic light beads glow when light falls on them.

CHEMISTRY

84. Soft drinks such as colas contain significant quantities of

(a) Caffeine (b) Nicotine
(c) Tannin (d) Renin

Ans. (a) : Caffeine is a xanthine alkaloid compound that acts as a stimulant in humans. Caffeine is sometimes called as guaranine, when found in guarana, mateine when found in mate and theine when found in tea. It is found in the leaves and beans of the coffee plant, in tea, yerba mate and guarana berries and in small quantities in cocoa.

85. Aluminium surfaces are often 'anodized'. This means the deposition of a layer of

(a) Chromium oxide (b) Aluminium oxide
(c) Nickel oxide (d) Zinc oxide

Ans. (b) : Anodizing is an electrolytic passivation process used to increase the thickness of the natural oxide layer on the surface of metal parts. The process is called "anodizing" because the part to be treated forms the anode electrode of an electrical circuit. Anodizing increases corrosion and withstand resistance. The anodized aluminium layer is grown by passing a direct current through an electrolytic solution, with the aluminium object serving as the anode. The current releases hydrogen at the cathode and oxygen at the surface of the aluminium anode, creating a buildup of aluminium oxide layer.

86. Consider the following statements:
Hard water is not suitable for
1. Drinking
2. Washing clothes with soap
3. Use in boilers
4. Irrigating crops
Which of these statements are correct?

(a) 1 and 3 (b) 2 and 3
(c) 1, 2 and 4 (d) 1, 2, 3 and 4

Ans. (d) : Hardness of water is due to presence of chlorides, sulphates and nitrates of calcium and magnesium. Hard water cannot be used for washing purposes as soap do not form lather with hard water easily which results in to wastage of soap in laundries. Moreover use of hard water in industries results in to scale formation in inner layers of boilers on which insoluble calcium and magnesium carbonates gets deposited and being a bad conductor of heat result in to wastage of energy. Hard water is not fit for drinking purpose as it hinders ionic imbalance in body. Repeated irrigation of crops by hard water increases calcium and magnesium ions in soil which increases soil alkalinity.

87. Match List I (Industrial processes) with List II (Industry with which associated) and select the correct answer using the codes given below:

List I	List II
A. Cracking	1. Rubber
B. Smelting	2. Petroleum
C. Hydrogenation	3. Copper
D. Vulcanization	4. Edible Fats

	A	B	C	D
(a)	3	2	1	4
(b)	2	3	4	1
(c)	2	3	1	4
(d)	3	2	4	1

Ans. (b) : Cracking is the process whereby complex organic molecules such as kerosene or heavy hydrocarbons are broken down into simpler molecules such as light hydrocarbons, by the breaking of carbon-carbon bonds. **Smelting** is a form of extractive metallurgy. Its main use is to produce a metal from its ore. In this process, the oxide ore is reduced to free metal by using reducing agents like coke or charcoal. Copper is also obtained from its oxide by this method in free state. **Hydrogenation** is a reduction reaction which results in an addition of hydrogen (usually as H2). If an organic compound is hydrogenated, it becomes more 'saturated'. Hydrogenation results in the conversion of liquid vegetable oils to solid (edible) fats.
Vulcanization is a chemical process for converting rubber into more durable materials via the addition of sulphur. Sulphur modify the rubber by forming crosslinks (bridges) between individual polymer chains. Vulcanized material is less sticky and has superior mechanical properties.

88. Assertion A : Large cold storage plants use ammonia as refrigerant while domestic refrigerators use chlorofluoro-carbons.

Reason R : Ammonia can be liquified at ambient temperatures at low pressures.

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Ans. (a) : Both A and R are individually true and R is the correct explanation of A.
Large cold storage plants use ammonia as refrigerant while domestic refrigerators use chlorofluorocarbons because ammonia could only be liquefied at ambient temperature and low pressure, which is not possible at the domestic refrigerators. That is why chlorofluorocarbons (CFC) is used in domestic refrigerators as a coolant though it is harmful to the environment. Hence, both A and R are individually true and R is the correct explanation of A.